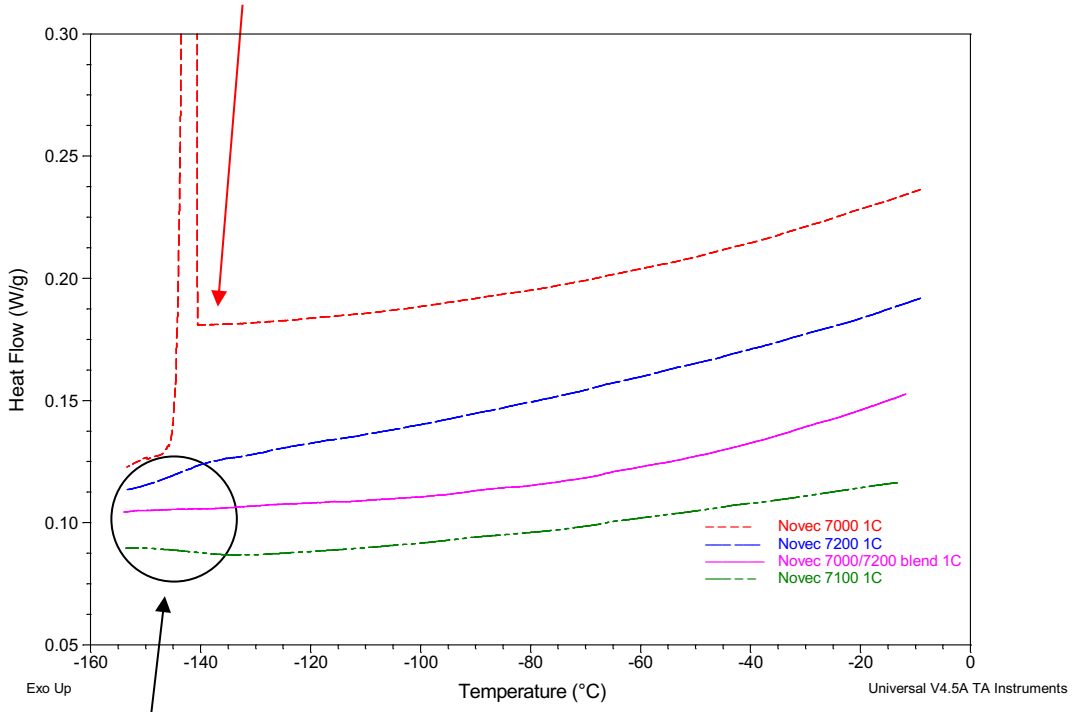
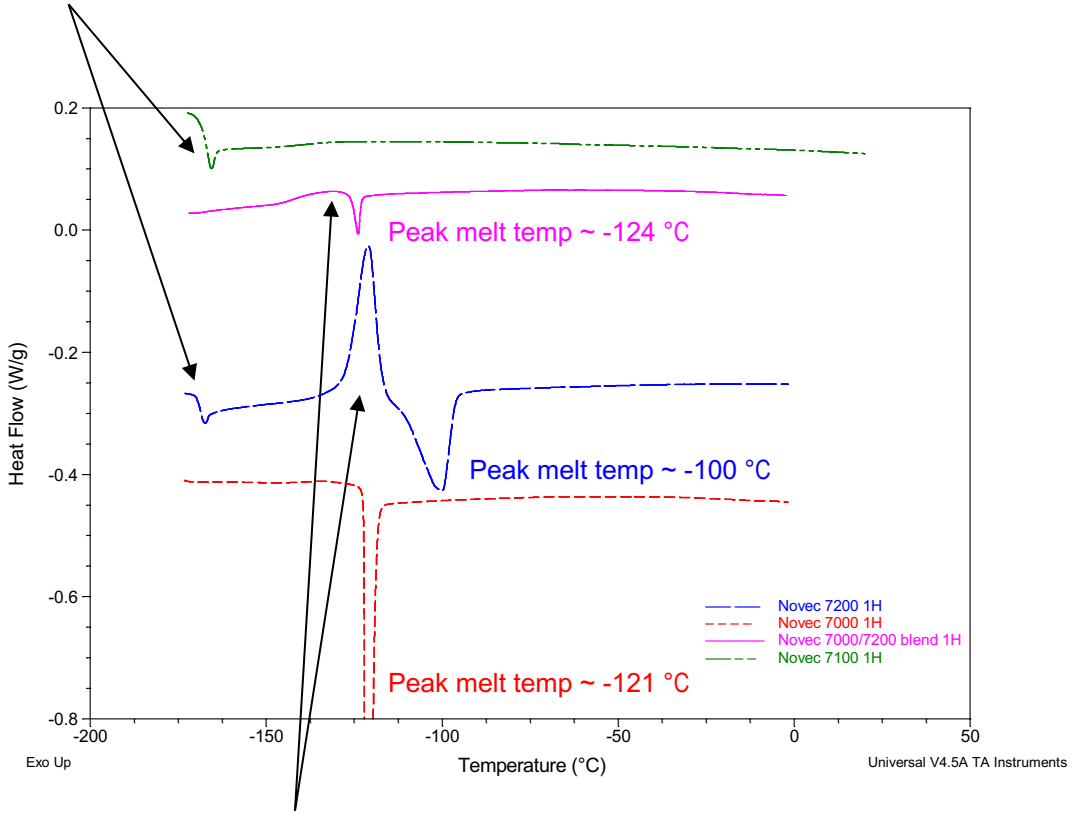
Novec 7072: Preliminary DSC Results

Rapid crystallization



Subtle baseline shifts suggest onset of nuclei/crystal formation

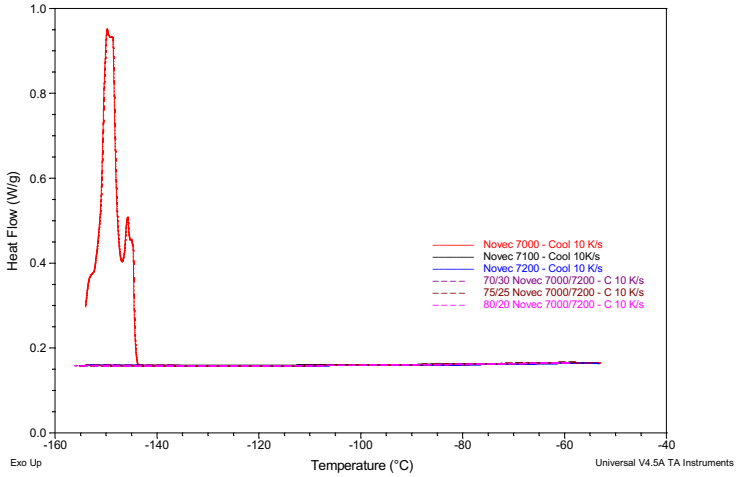
Glass transition

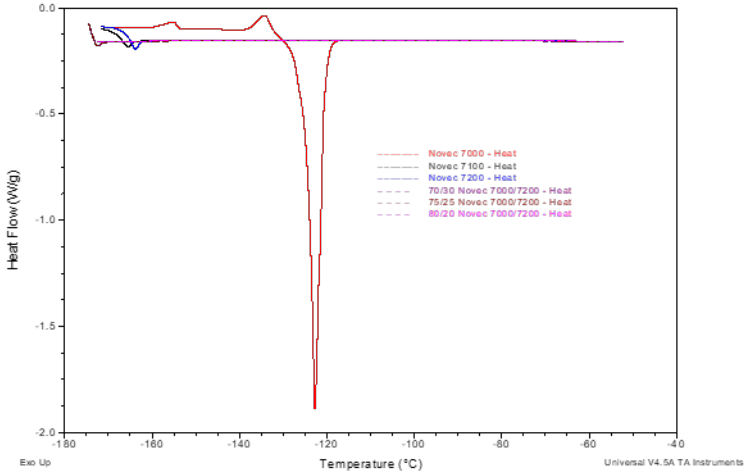


Cold crystallization

Novec 7072: DSC Results Summary:

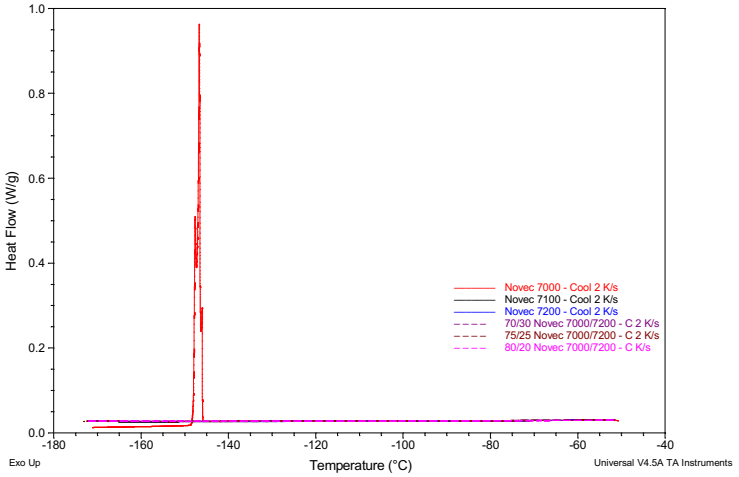
Cooling at 10 K/s

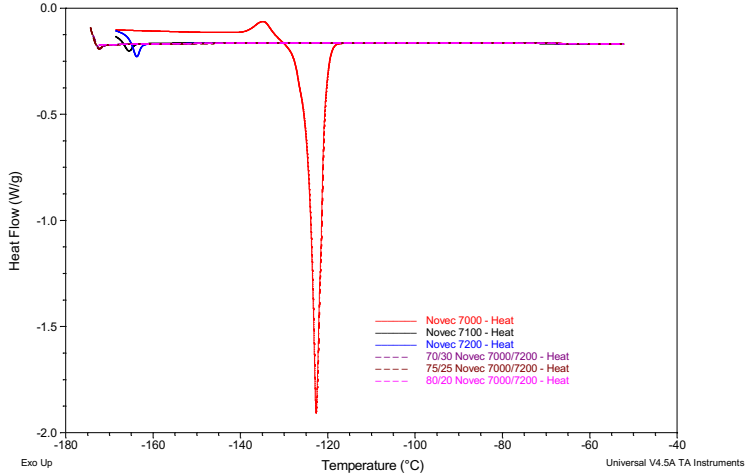




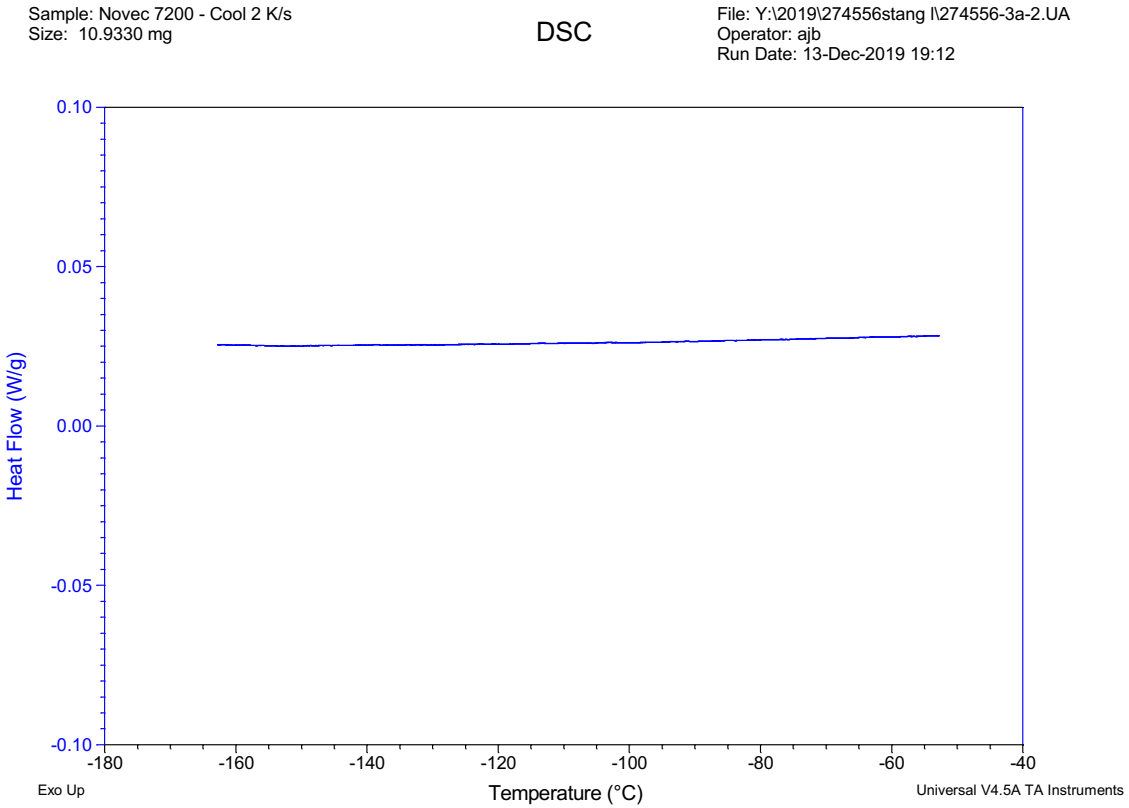
* Only pure **Novec 7000** shows any signs of crystallization cooling and melting on heating

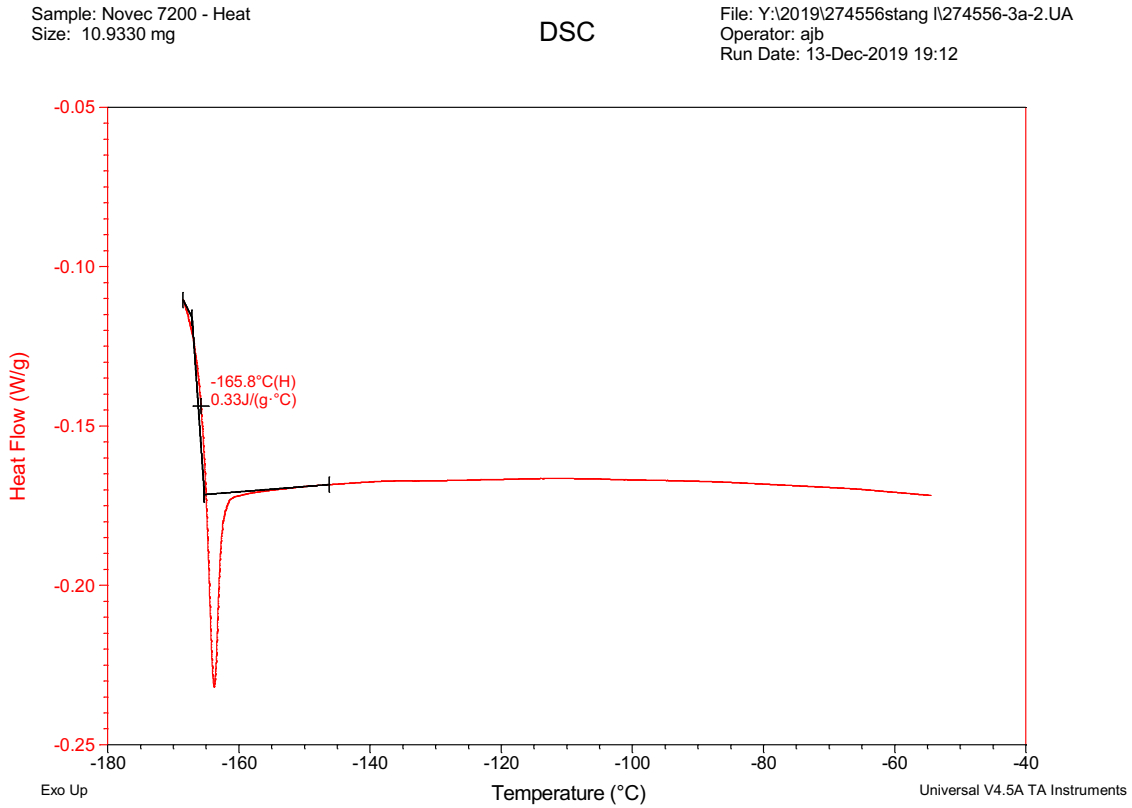
Cooling at 2 K/s



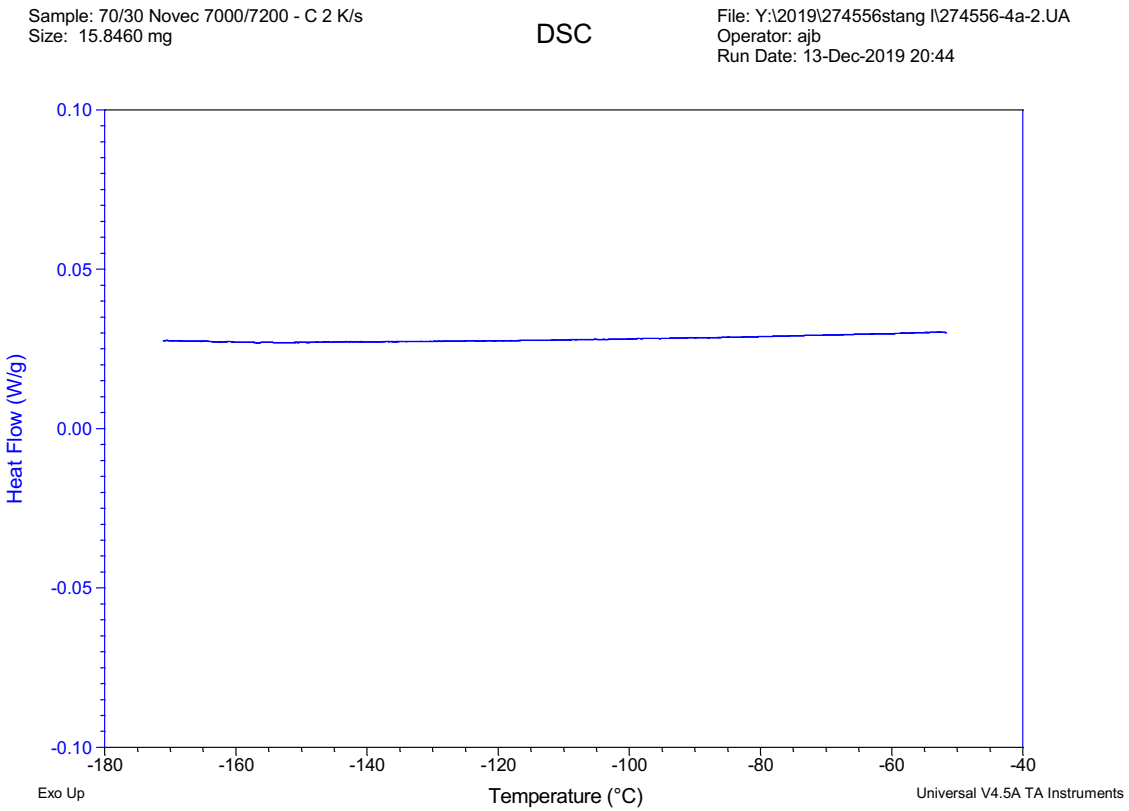


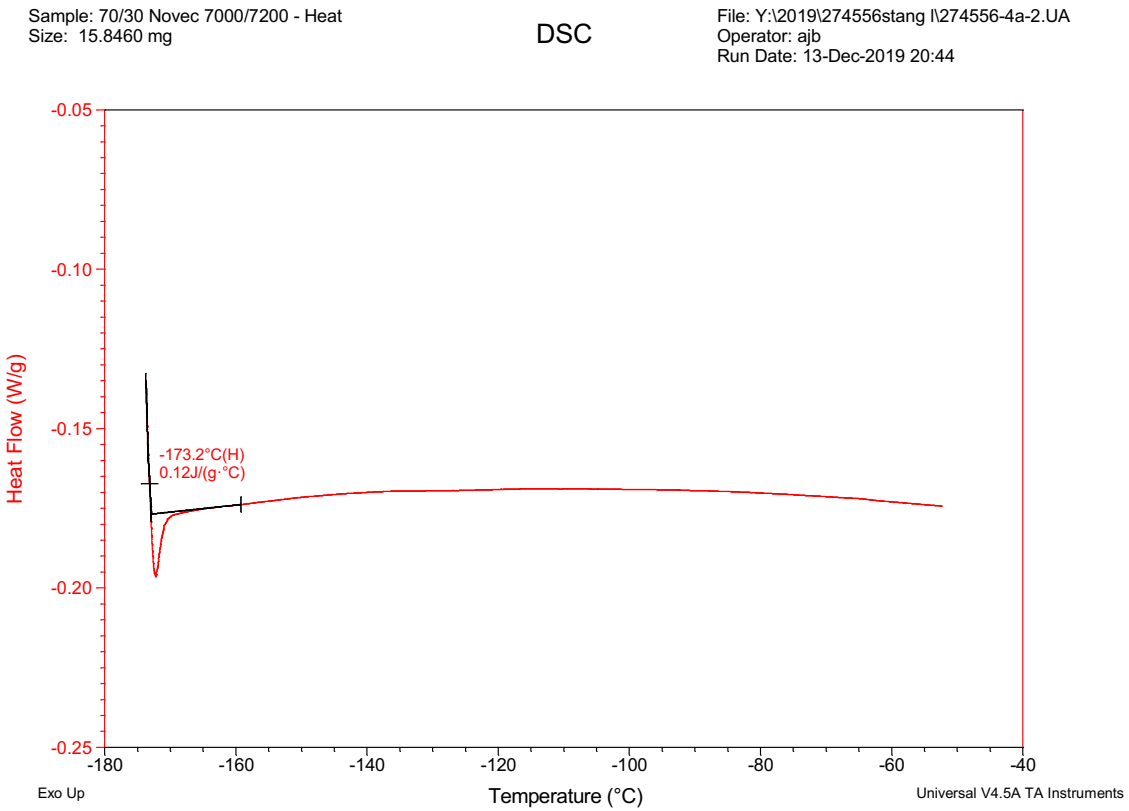
Novec 7200


Sample: Novec 7200 - Cool 2 K/s
Size: 10.9330 mg
DSC
File: Y:\2019\274556stang I\274556-3a-2.UA
Operator: ajb
Run Date: 13-Dec-2019 19:12
Heat Flow (W/g)
0.10
0.05
0.00
-0.05
-0.10
-180
-160
-140
-120
-100
-80
-60
-40
Exo Up
Temperature (°C)
Universal V4.5A TA Instruments

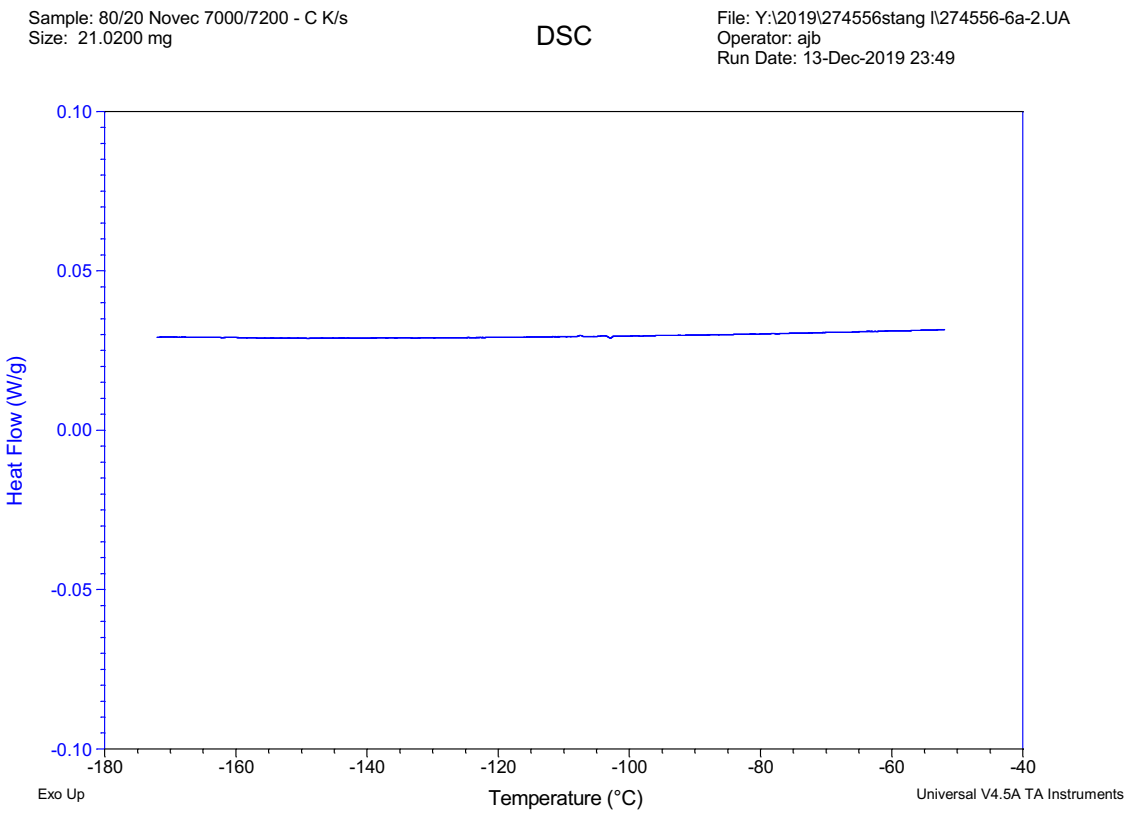

Sample: Novec 7200 - Heat
Size: 10.9330 mg
DSC
File: Y:\2019\274556stang I\274556-3a-2.UA
Operator: ajb
Run Date: 13-Dec-2019 19:12
Heat Flow (W/g)
-0.05
-0.10
-0.15
-0.20
-0.25
-165.8°C(H)
0.33J/(g·°C)
-180
-160
-140
-120
-100
-80
-60
-40
Exo Up
Temperature (°C)
Universal V4.5A TA Instruments

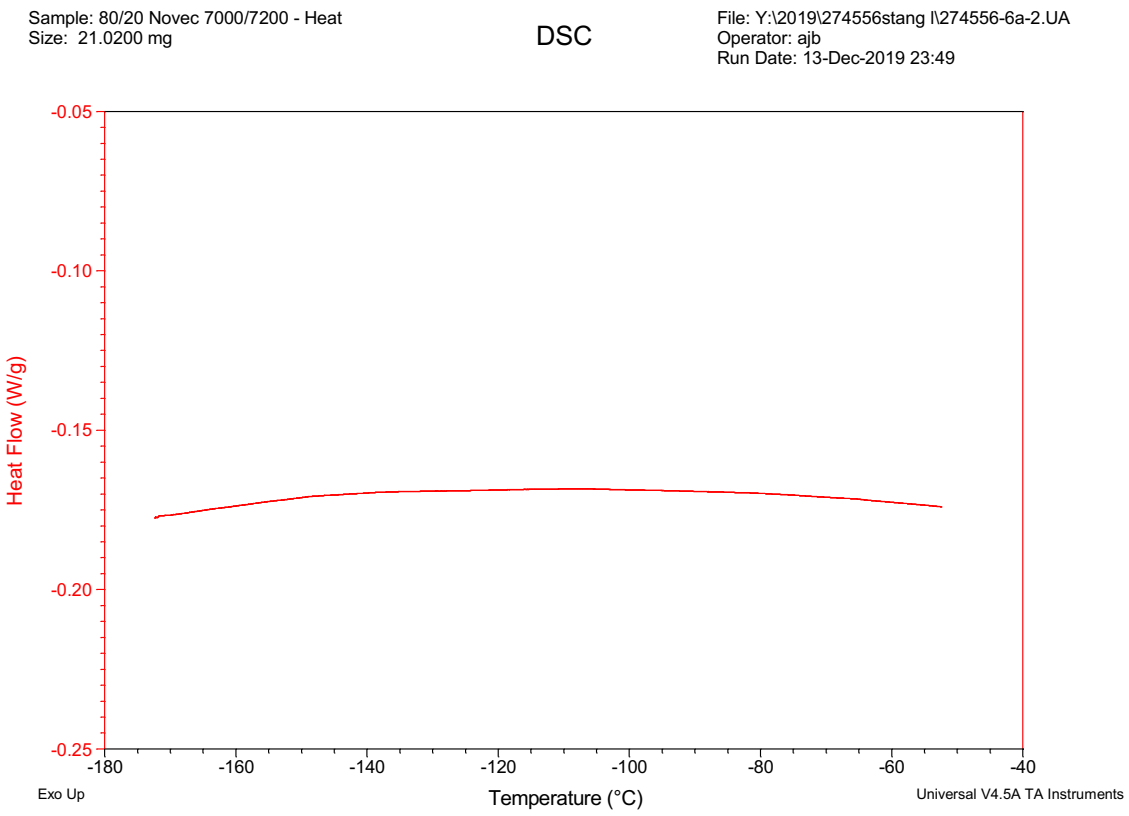
70/30 Novec 7000/7200 blend





80/20 Novec 7000/7200 blend

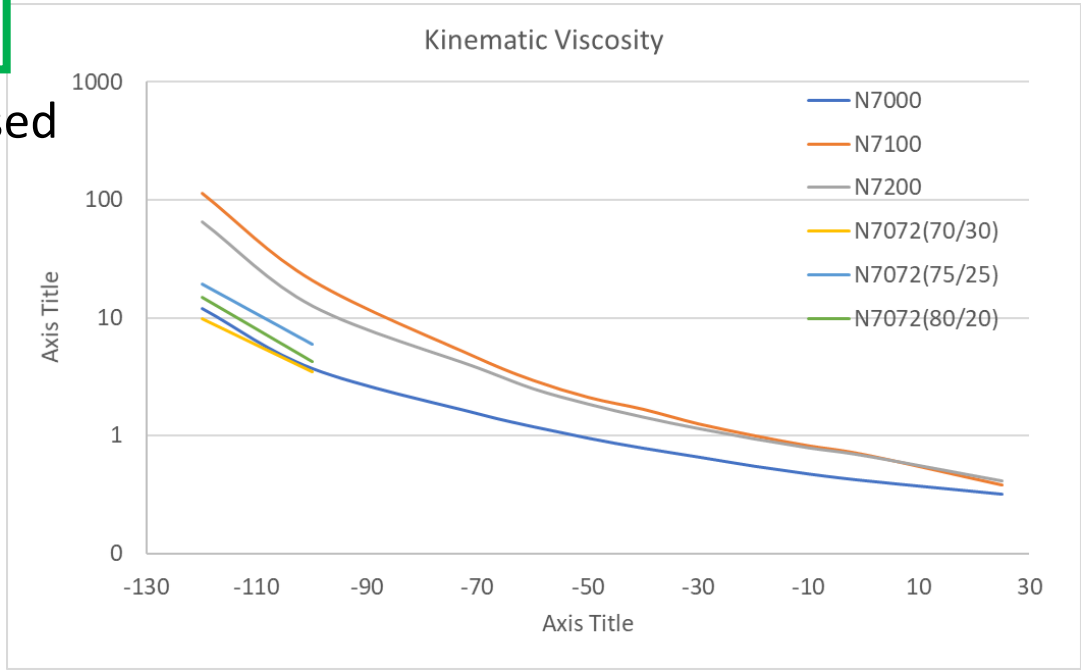




Viscosity

TdegC	N7000	N7100	N7200	N7072(70/30)	N7072(75/25)	N7072(80/20)
25	0.32	0.38	0.41			
0	0.42	0.69	0.67			
-10	0.47	0.82	0.78			
-20	0.55	1.00	0.93			
-30	0.66	1.25	1.14			
-40	0.78	1.67	1.42			
-50	0.95	2.11	1.84			
-60	1.19	2.94	2.48			
-70	1.53	4.53	3.72			
-100	3.69	20.63	12.47	3.50	5.96	4.29
-120	11.87	113.35	64.47	9.88	19.12	14.95

Both N7100 and N7200 suppress the PP of N7000 but N7200 has lower viscosities at low temps.

Blend proposed For nEXO

Test data for 70/30 blend is unusual and should be retested